

Generated Question Paper

Q1. The term 'species' is a term used to describe a species. [2 marks]



Fig. 6.1

Figure: potato_varieties.png

Q2. The average photosynthesis rate of soybean plants was measured in two plots with different CO₂ concentrations, 370 ppm and 550 ppm. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

The atmospheric carbon dioxide concentrations in the two plots were kept at:

- 370 ppm, which is similar to the current atmospheric carbon dioxide concentration
- 550 ppm, which is a possible future atmospheric carbon dioxide concentration.

When the soybean plants were fully grown, scientists calculated the average rates of photosynthesis at regular intervals from 04:00 to 22:00 for both plots.

The results are shown in Fig. 2.1.

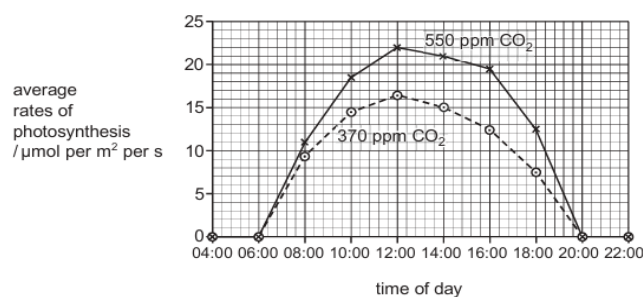


Fig. 2.1

Figure: photosynthesis_rate_graph.png

Q3. (a) State the hormone released from the testes. [1 marks]

Q4. The process of converting ammonium ions into nitrate ions is called the nitrate reaction. [1 marks]

Q5. (b) (i) Name the source of energy used by organism A. [1 marks]

Q6. (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q7. The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. [2 marks]

Q8. noindent_____ (ii) State a conclusion for the results shown in Table [1 marks]

Q9. The retina is a part of the retina. What is the retina's function? [3 marks]

Q10. (f) Describe the process by which a fertilised egg cell is fertilised. [3 marks]

Q11. The advantage of using a pyramid of energy is that it is more efficient. [2 marks]

Q12. (b) State the organs that make up the central nervous system. [1 marks]

Q13. Plants store starch in their leaves. [2 marks]

Q14. Plants growing in Europe begincenter begintabularcc toprule Plant Number & Leaf Area (cm\$2\$) midrule 1 & 310 2 & 335 3 & 390 4 & 348 5 & 365 midrule Mean & 350 bottomrule endtabular endcenter (i) Using the data in Table 5.2, calculate the mean leaf area for plants growing in North America. Give your answer as a whole number and include the unit. [2 marks]

Q15. (ii) Name the process that causes gas bubbles to form in the dough. [1 marks]

Q16. The testes are a part of the human body. Meiosis is the process of dividing the testes into two parts. [3 marks]

One has been completed for you.

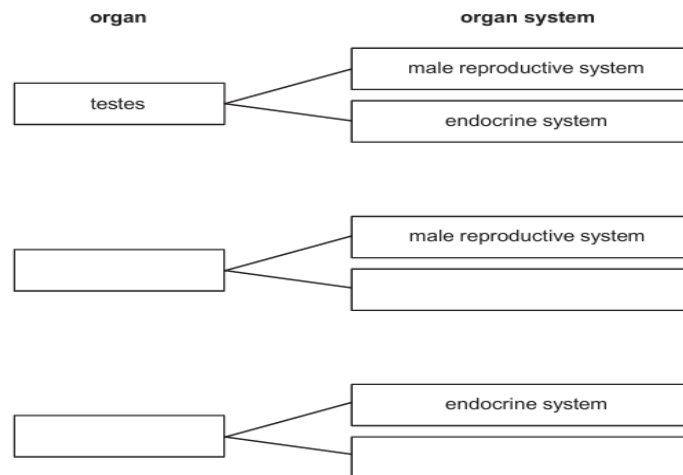


Fig. 5.1

Figure: organs_two_systems.png

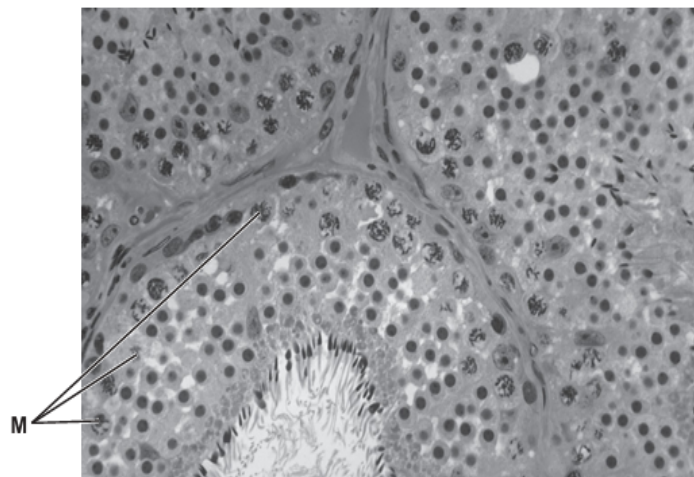


Fig. 5.2

Figure: mammalian_testis_micrograph.png

Q17. A sperm cell is a cell that divides into two individual cells. A sperm cell is a cell that divides into two individual cells. A sperm cell is a cell that divides into two individual cells. [3 marks]



Fig. 5.3

Figure: sperm_cell_micrograph.png